

Prabhat Kumar Singh . Jarvis Consulting

Cloud and Data Engineer with over two years of experience building scalable data pipelines, cloud-based data solutions, machine learning workflows, and enterprise integrations for Fortune 500 clients. Experienced with Python, SQL, PySpark, APIs, AWS services including S3, Glue, Athena, Lambda, and SageMaker, and Azure technologies including Data Factory, Databricks, Synapse, and ADLS Gen2. Built solutions supporting promotion analytics, forecasting, automation, and enterprise data migration. Also developed hands-on AI projects using LangChain, embeddings, FAISS vector search, and retrieval-augmented generation. Interested in building reliable cloud, platform, automation, and AI-enabled solutions that solve measurable business problems.

Skills

Proficient: Python, AWS, Azure, Databricks, Pyspark, Sql, Power BI

Competent: REST APIs, Docker, PostgreSQL, Data Modeling, ETL Pipeline Development, Machine Learning(Regression/Clustering)

Familiar: Hadoop/HDFS, Retrieval-Augmented Generation (RAG), Java, Linux/Unix, Financial Reporting/Financial Forecasting

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_Singh

Linux Cluster Resource Monitoring App [GitHub]: Designed and implemented a lightweight Linux cluster monitoring system using Bash scripts and crontab to collect hardware specifications and real-time CPU and memory utilization metrics. Stored collected telemetry in a PostgreSQL database running inside Docker and developed SQL-based analytics queries to support capacity planning and performance monitoring. Automated scheduling and deployment to enable reliable, low-overhead monitoring across distributed environments.

PostgreSQL Club Booking Database System [GitHub]: Designed and implemented a normalized relational database system for a recreational club booking platform using PostgreSQL. Created schemas, constraints, and relationships using DDL statements and developed advanced SQL queries leveraging joins, aggregations, subqueries, and window functions to generate operational insights. Containerized the database environment using Docker and automated deployment with Bash scripts to improve portability and reproducibility across environments.

Retail Data Wrangling and Analytics [GitHub]: Developed a retail analytics solution to analyze customer purchasing behavior using Python, Pandas, SQLAlchemy, PostgreSQL, and Jupyter Notebook. Performed data cleaning, transformation, and exploratory analysis on transactional datasets and applied RFM segmentation and monthly sales trend analysis to generate customer-level insights. Delivered analytics-ready datasets and data-driven marketing recommendations to support customer retention strategies and campaign targeting.

Java Grep Application [GitHub]: Developed a command-line text search application in Java 8 that replicates core Linux grep functionality by recursively scanning directories for pattern-based matches using regular expressions. Implemented a functional programming approach using Lambda expressions and the Stream API to improve processing efficiency and memory usage through lazy evaluation. Integrated SLF4J for logging and Maven for dependency management, and packaged the application within a Dockerized environment to enable portable execution across systems while supporting scalable processing of large directory structures.

PySpark Data Analysis [GitHub]: Built distributed data analytics pipelines using PySpark across Databricks and Zeppelin notebook environments to analyze large-scale retail and economic datasets. In Databricks, ingested retail transaction data from PostgreSQL via JDBC and implemented end-to-end analytics with PySpark DataFrames, including monthly sales trends, placed vs. canceled orders, sales growth rates, active users, new vs. existing customer analysis, and RFM-based segmentation. In Zeppelin, analyzed World Development Indicators (WDI) data using PySpark with Hive-backed tables, applying transformations, joins, filtering, and aggregations to strengthen distributed processing workflows. Delivered scalable, business-focused insights into customer behavior and sales performance using structured Spark DataFrame APIs.

ETL pipeline for Financial Data [GitHub]: Financial Fraud Analytics ETL Pipeline [GitHub]: Designed and implemented an end-to-end fraud analytics data pipeline on Azure Databricks using Medallion Architecture (Bronze, Silver, Gold) to ingest, clean, enrich, and analyze multi-source financial transaction datasets. Integrated Azure SQL Database,

Azure Data Lake Storage Gen2, PySpark, and JDBC to process transactions, card data, user profiles, merchant category codes, and fraud labels into analytics-ready Delta tables. Built curated Gold-layer datasets to support fraud trend monitoring, high-risk user identification, merchant-category fraud rate analysis, daily fraud loss tracking, and behavioral pattern changes before and after fraud events. Developed interactive Databricks dashboards and scheduled Databricks Jobs to orchestrate automated pipeline execution from ingestion to reporting refresh, enabling scalable and production-style analytics workflows.

COBOL Student Registration System [GitHub]: Built a COBOL-based Student Registration System using Open-CobolIDE and GnuCOBOL to simulate mainframe-style indexed file processing. Developed a menu-driven application to generate a student master file, insert new student records, update existing records, delete records, and query students by ID or insertion date. Implemented course-wise report generation using COBOL sorting, sequential file handling, indexed file access, record layouts, and structured procedural logic. Strengthened practical understanding of COBOL fundamentals, copybooks, file control entries, working-storage design, CRUD operations, and report generation in a mainframe-style business application.

Highlighted Projects

LangChain-FinanceGPT [GitHub]: Developed an AI-powered financial question-answering system using LangChain, OpenAI embeddings, FAISS vector search, and Flask to enable natural language querying of company annual reports. Implemented document ingestion pipelines to process large financial PDFs into vector embeddings and built a retrieval-augmented generation workflow to deliver context-aware responses for investment research and financial analysis use cases.

Kubernetes for Data Engineering [GitHub]: Designed and deployed a containerized data engineering workflow using Docker and Kubernetes to orchestrate scalable data processing services across distributed environments. Implemented infrastructure configuration for automated service deployment, container lifecycle management, and environment portability to support reliable execution of analytics workloads in cloud-native architectures.

Professional Experiences

Data Engineer, Jarvis (2026-present): Contributed as a Data Integration Engineer and Developer on an engagement for Arterra Wines. Developed and enhanced Python workflows to transform and migrate customer, address, payment-token, product, and subscription data from WineDirect into Salesforce Commerce Cloud, Adyen, and Ordergroove. Implemented data mapping, validation, deduplication, and error-handling logic to improve migration accuracy and resolve integration issues. On the development side, contributed to the Salesforce fulfillment site using Lightning Web Components and Apex, including enhancements to the pick-ticket queue and location-based order filtering. Collaborated with business and technical teams to understand requirements, test integrations, troubleshoot issues, and deliver reliable migration and fulfillment solutions.

Data Engineer, Decision Point Analytics (Feb 2021-Dec 2022): Designed and supported AWS-based data engineering and Promotion Analytics solutions for Fortune 500 retail and CPG clients. Built Python, SQL, and PySpark data pipelines to process large volumes of historical sales, pricing, product, and promotional data using AWS services including S3, Glue, Athena, Lambda, and SageMaker. Developed transformation, validation, and automation workflows that prepared analytics-ready datasets for promotion measurement and machine learning use cases. The broader Promotion Analytics initiative contributed to more than \$15 million in revenue uplift and improved promotional ROI by 12%. Collaborated with business and technical teams to translate analytical requirements into scalable cloud-based solutions and reliable data workflows.

Education

SRM University (2016-2020), Bachelor of Computer Sciences, Computer Science - First Class with Distinction

Georgian College (2023-2024), Postgraduate Certificate in Artificial Intelligence, School of Information Technology - Dean's List

Miscellaneous

- Microsoft Certified: Azure Data Fundamentals
- Databricks Certified Data Engineer Associate
- IBM Data Science Professional Certificate
- Machine Learning with Python (IBM)

- Databases and SQL for Data Science with Python (IBM)
- Introduction to the Internet of Things (IoT)- Curtin University